

Pathwise

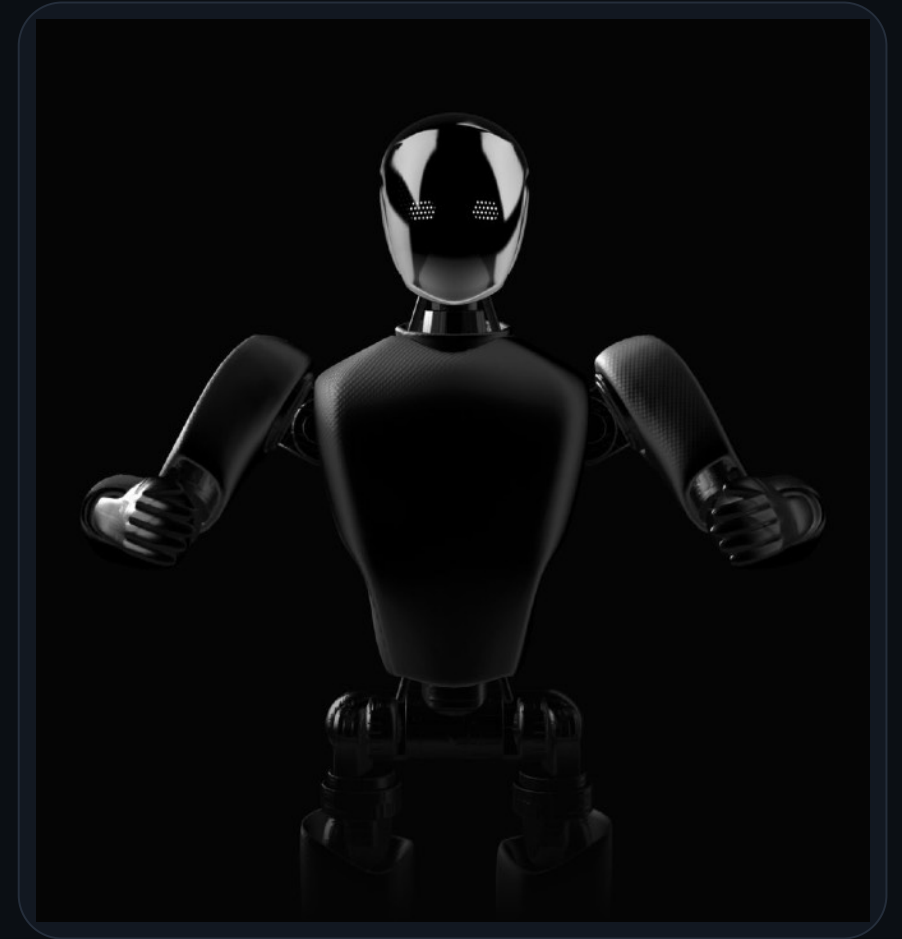
AI-powered personalized learning path recommender

A learning path you can verify.

trypathwise.vercel.app

PROBLEM STATEMENT

Design and prototype an AI-powered solution that delivers personalized learning experiences based on an individual's needs, interests, learning patterns and goals.



The problem isn't the brief. It's 11pm and a browser tab.



Roadmaps everywhere

"40 tutorials, no order"



No idea what's next

"which course actually builds
on this?"



Progress is a video bar

"% watched, not skills gained"



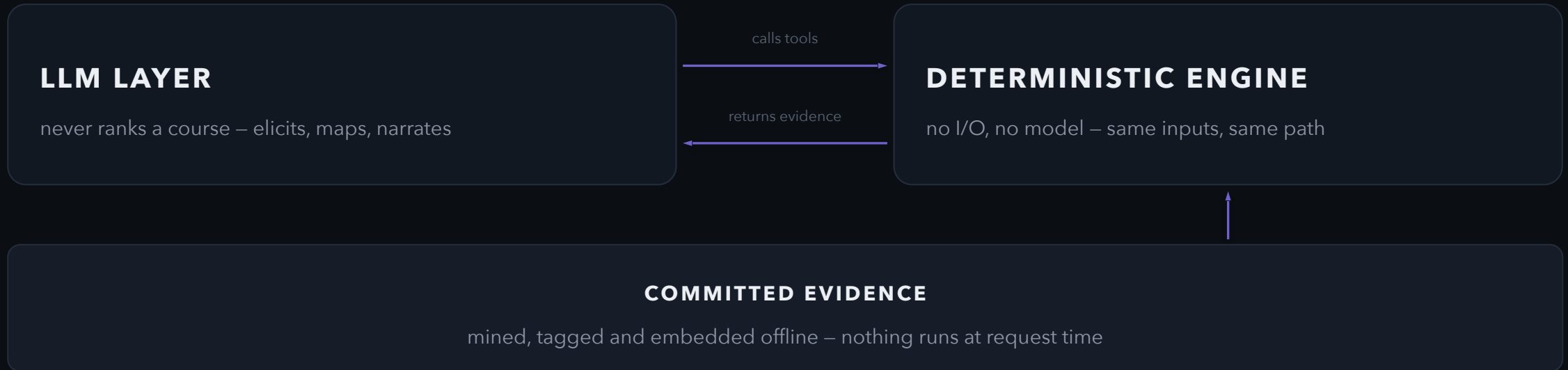
Advice, unverified

"a guess dressed up as a
recommendation"

"Everyone gets advice. No one gets a reason to trust it."

the gap Pathwise is built to close

The LLM narrates. The engine decides.



Every recommendation is traceable to arithmetic.

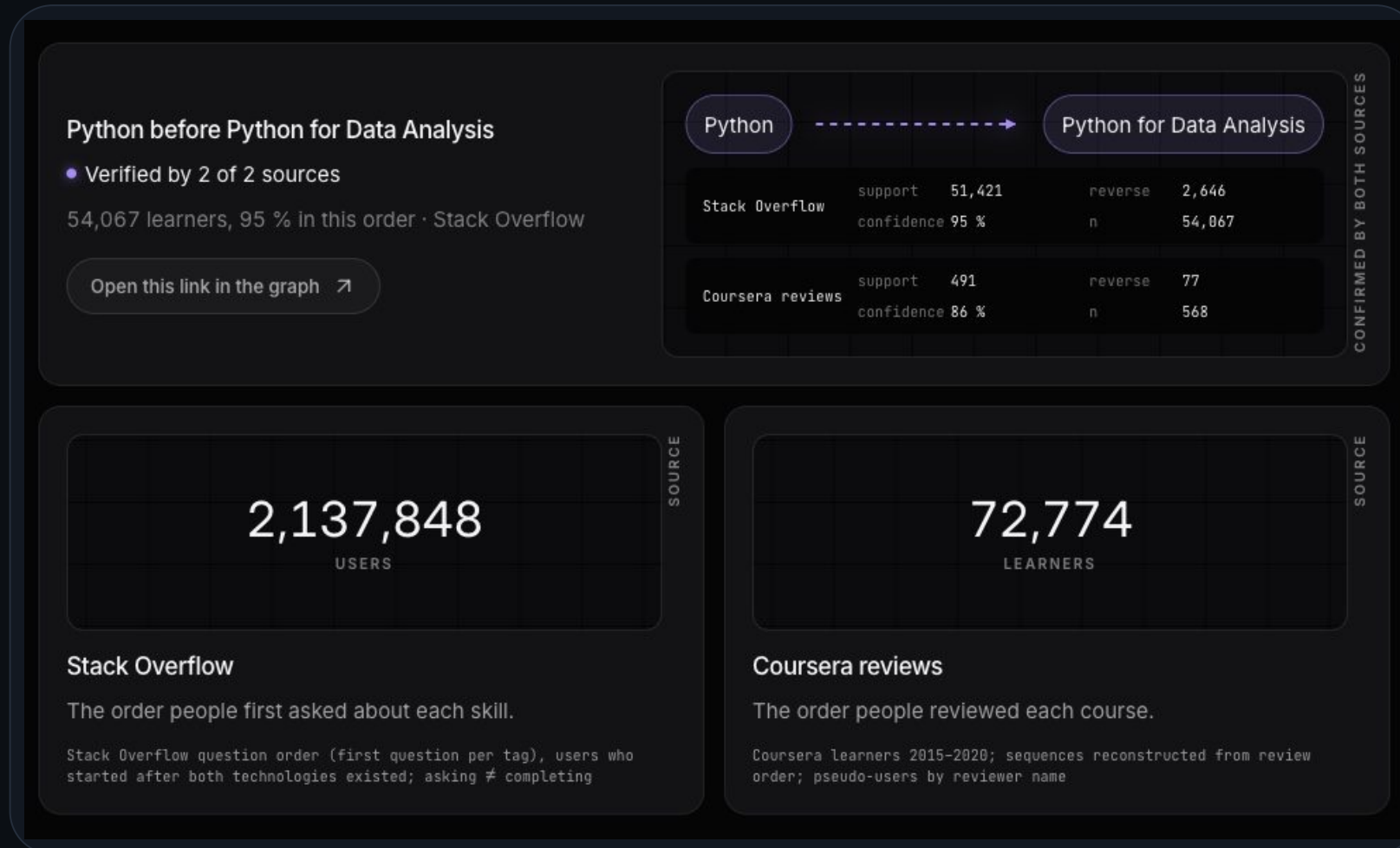
Open one file. Reconstruct by hand why a course is item three.

Precomputed offline. Deterministic at runtime.

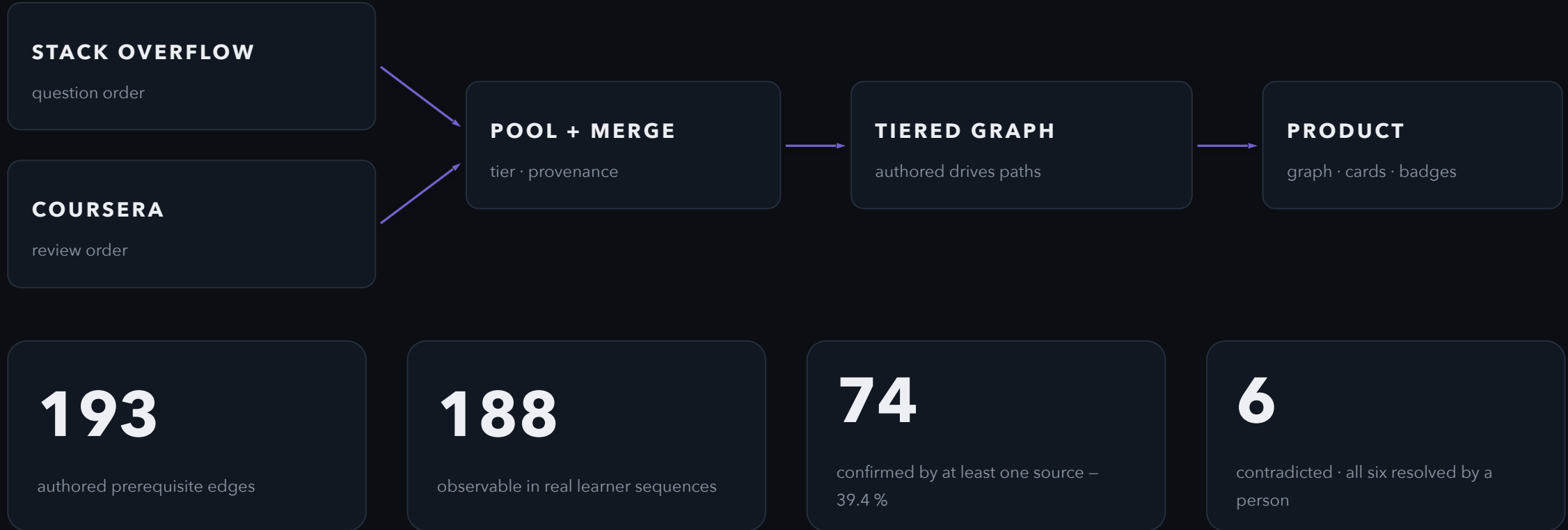


One-way rule: the LLM calls the engine as tools. The engine never calls the LLM.

We checked the map against real learners.



Evidence annotates the map. Humans decide.



0 novel edges promoted. Nothing automatic flips an edge.

Three more measurements, reported as they came.

MiniLM kept

gte-base led P@1 0.803 vs 0.776 – every 95 % interval straddled zero

61.7 %

of 60 narrations carried an unsupported claim ·
25.0 % an invented fact or a misstated field

4.78 %

most of 983 path items any single weight moves
at ± 25 % · no phase order flipped

The challenger won the bake-off and we kept the incumbent, because the win was not clear.

The tools are the engine.

• Reading your profile

Marked done — that puts Command Line & Linux Basics at level 1 on your profile. Networking is next: three hours, and it explains what actually happens when a page loads.

Started the big JavaScript course. It's moving fast.

• It is a long one — 69 hours. If it assumes things you don't have yet, say so and I'll rework the path rather than have you push through.

The JavaScript course is too hard — it assumes I already know things about CSS layout and the DOM that I don't.

• Replanning your path

Understood. I've lowered what I assumed you hold in CSS and JavaScript and redone the path: the freeCodeCamp Responsive Web Design certification and a small GitHub Pages site come in before any JavaScript course, and the two items that leaned on those skills are out for now. The diff on your path shows exactly what moved and why.

STRUCTURED OUTPUTS

`messages.parse()` against Zod schemas — the skill id is an enum, so the model cannot emit a skill that does not exist

THE TOOLS ARE ENGINE FUNCTIONS

`get_profile` · `apply_profile_ops` · `generate_path` · `explain_item` · `search_catalog` — capped at six iterations per turn

PROMPT CACHING

frozen system prompts; cache reads verified in usage

DEGRADATION BY DESIGN

past a token cap the deterministic core keeps working

Six asks, all shipped.



- 01 Conversational intake – Nova, over an SSE chat stream
- 02 Learner profiling – levels carry stated / inferred / assessed
- 03 Real resources – 307 courses, 36 projects, 27 assessments
- 04 Path generator – 159-skill DAG, topological order, milestones
- 05 Explanation – evidence card, narration, provenance popover
- 06 Dashboard – progress, skills, timeline, next best action
- + Adaptation – feedback becomes a visible diff, with its cause

The diff, and the proof.

● PATH UPDATED

v4

8 changes

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Added MDN Learn Web Development, Frontend Mentor Challenges, The Modern JavaScript Tutorial, Testing Library Documentation and App Ideas Collection and removed Learn Accessibility, The Complete JavaScript Course: From Zero to Expert and Getting Started · Jest, and reordered the rest, because you found it too hard.

ADDED 5

REMOVED 3

+

MDN Learn Web Development

Rebuilds what The Complete JavaScript Course: From Zero to Expert assumed you knew — closes HTML, CSS, JavaScript...

+

Frontend Mentor Challenges

Rebuilds what The Complete JavaScript Course: From Zero to Expert assumed you knew — closes CSS, Responsive Design

+

The Modern JavaScript Tutorial

Rebuilds what The Complete JavaScript Course:

-

Learn Accessibility

Made room for remediation first

-

The Complete JavaScript Course: From Zero to Expert

Made room for remediation first

-

Getting Started · Jest

Made room for remediation first

feedback → replan → a visible diff

SQL

Python

Command Line & Git

TypeScript

Algorithms

Advanced JS

MySQL

Data Modeling

ETL & Data Pipelines

Probability

Shell Scripting

Security Fundamentals

Node.js

Vue.js

Tableau

Regression Analysis

Data Lakehouse

Data Warehousing

Regression Analysis

Apache Airflow

Python for Data Science

Statistics Fundamentals

Power BI

Tableau

Machine Learning & Data Science

Modern JavaScript

Cloud

Big Data & Analytics

Advanced ML

Traditional BI Tools

Concurrency & Parallelism

Design Patterns

Robot Framework

Performance

Python before Python for Data Analysis

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● Verified by 2 of 2 sources

54,067 learners, 95 % in this order · Stack Overflow

^ hide details

Confirmed by both sources

Stack Overflow

question order · CC BY-SA 4.0

took Python... took Python... in this order n

51,421 2,646 95 % 54,067

TAGS python, python-2.7, python-3.x, dask, numpy, numpy-ndarray, pandas, pandas-groupby

Stack Overflow question order (first question per tag), users who started after both technologies existed, asking # completing

Coursera reviews

review order · Kaggle corpus 2015-2020

took Python... took Python... in this order n

explain → evidence → provenance

What was actually hard.

370

real items curated by hand, each
with a live URL

0.9149

reviewer-vs-model Jaccard on
course→skill tags · gate 0.85 · one
delegated reviewer

8,260

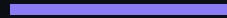
pairs left of 8,709 once the cohort-
bias filter is applied

\$1.8847

offline tagging over 273 calls ·
\$1.07 for the groundedness pass

Accepted knowingly – no users yet · the sources measure asking and reviewing, not mastery · the Coursera corpus ends in 2020

Kept fresh by machines, kept true by people.



trypathwise.vercel.app

github.com/iAnmolAgarwal/pathwise

Stack Overflow data under CC BY-SA 4.0 · Coursera review order via the Kaggle corpus

The learner-sequence mining method, its measured validation (successes and failures shown side by side) and the “neither graph is strictly superior – use both and say which” conclusion are Riyan Garg’s; Pathwise reproduces his method in the repository.